

INSTRUCTIONS TO THE CANDIDATES

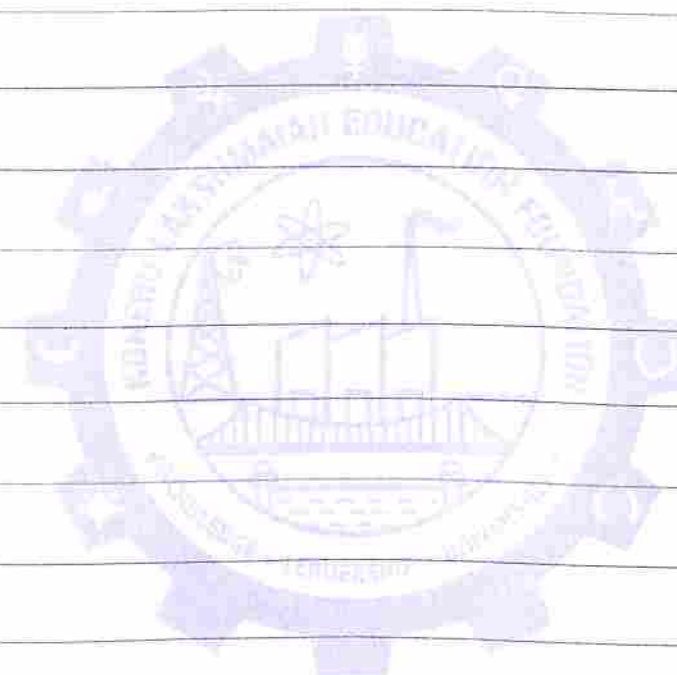
- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

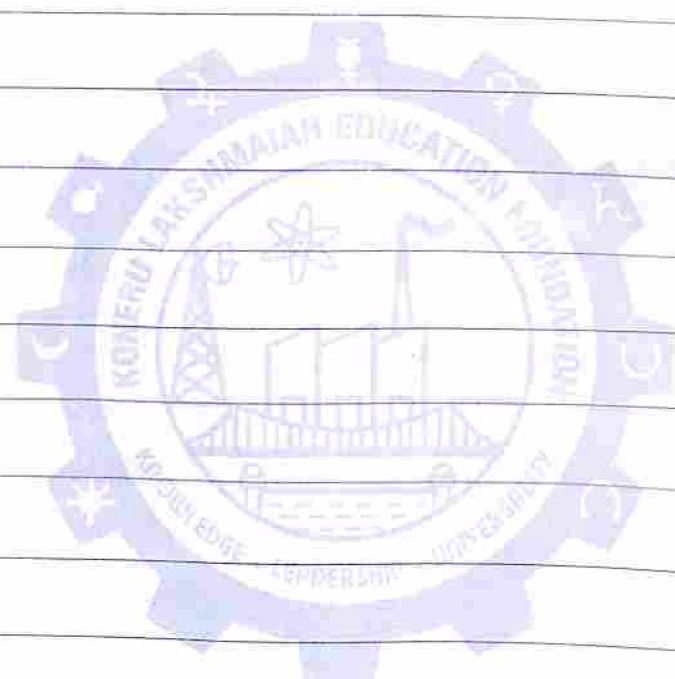
Non-adherence with this instruction shall result in candidate being booked under malpractice.

- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.



1 A

$$P(A) = 0.4$$

$$P(B) = 0.6$$

$$P(A \cap B) = 0.25$$

$$(i) P(A \cup B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.4 + 0.6 - 0.25$$

$$= 1.0 - 0.25$$

$$= \boxed{0.75}$$

$$(ii) P(\bar{A} \cap B)$$

$$P(\bar{A} \cap B) = P(B) - P(A \cap B)$$

$$= 0.6 - 0.25$$

$$= \boxed{0.35}$$

$$(iii) P(A \cap \bar{B})$$

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$= 0.4 - 0.25$$

$$= \boxed{0.15}$$

$$(iv) P(\bar{A} \cap \bar{B})$$

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$$

$$= 1 - P(A \cup B)$$

$$= 1 - 0.75$$

$$= \boxed{0.25}$$

1 B

i) Random Variables

- Discrete $\rightarrow \sum$ Sum
- Continuous $\rightarrow \int$ Integration

ii) Cumulative Distribution function.

x	0	1	2
$P(x)$	0.3	0.6	0.1

$$\sum P(x=x_i) = 1$$

1 C

Class $\rightarrow 40$

Cricket $\rightarrow 28 = P(C)$

Football $\rightarrow 12 = P(F)$

$$\begin{aligned}
 \text{either } C \text{ or } F &= P(C \cup F) \\
 &= P(C) + P(F) - P(C \cap F) \\
 &= 28 + 12 \\
 &= \frac{40}{40} = 1
 \end{aligned}$$

1 D

Correlation

6 types

- $\rightarrow$ Strong +ve
- $\rightarrow$ Strong -ve
- $\rightarrow$ Weak +ve
- $\rightarrow$ Weak -ve
- $\rightarrow$ Zero / None
- $\rightarrow$ Complex

3 methods.

- $\rightarrow$ Scatter
- $\rightarrow$ Karl Pearson.
- $\rightarrow$ Spearman.

1 E

Mean, Median, Mode

5 8 10 12 12 15 16 18 20

$$5 + 8 + 10 + 12 + 12 + 15 + 16 + 18 + 20 = 116$$

$$\text{Mean} = \frac{116}{9} \Rightarrow 12.88$$

Avg

$$\text{Median} = 12$$

middle

$$\text{Mode} = 12$$

most repeated.

1 F

Random Exponential variable

$$\lambda = 0.5$$

$$\text{mean} = \frac{1}{\lambda} = \frac{1}{0.5} = 2$$

2 A

Integrated circuit chip

defective etching $P(D) = 0.06$

Crack defect $P(C) = 0.03$

Both $P(D \cap C) = 0.02$

i) either D or C

$$\begin{aligned} \Rightarrow P(D \cup C) &= P(D) + P(C) - P(D \cap C) \\ &= 0.06 + 0.03 - 0.02 \\ &= \boxed{0.07} \end{aligned}$$

ii) neither defect

$\Rightarrow P(\bar{A})$

$$\begin{aligned} \Rightarrow P(\bar{D} \cap \bar{C}) &= P(\overline{D \cup C}) \\ &= 1 - P(D \cup C) \\ &= 1 - 0.07 = \boxed{0.93} \end{aligned}$$

iii) Only one defect.

$$\begin{aligned} \Rightarrow P(D \cap \bar{C}) + P(\bar{D} \cap C) \\ &= P(D) + P(C) - 2P(D \cap C) \\ &= 0.06 + 0.03 - 2(0.02) \\ &= 0.09 - 0.04 \\ &= \boxed{0.05} \end{aligned}$$

2 B

Poisson.

$$\lambda = 2$$

$$\frac{e^{-\lambda} \lambda^x}{x!}$$

i) $x = 5$ $t = 2$

$$\begin{aligned} p &= \frac{e^{-2(2)} \cdot (2(2))^5}{5!} \quad e^{-4} = 0.018 \\ &= \frac{e^{-4} \cdot 128}{120} = \frac{0.018 \cdot 128}{15} = \frac{2.304}{15} \end{aligned}$$

ii) $x \geq 3$ $t = 4$

$$\begin{aligned} &= 1 - p(x < 3) \\ &= 1 - (p(x=0) + p(x=1) + p(x=2)) \end{aligned}$$

$$x=0 \quad t=4$$

$$\frac{e^{-2(4)} (2(4))^0}{0!} = 0$$

$$e^{-8} = 0.0003$$

$$x=1 \quad t=4$$

$$\frac{e^{-2(4)} (2(4))^1}{1!} = \frac{e^{-8} (8)}{1} = \frac{0.0003 \cdot 8}{1} = 0.0024$$

$$x=2 \quad t=4$$

$$\frac{e^{-2(4)} (2(4))^2}{2!} = \frac{e^{-8} (16)}{2} = \frac{0.0003 \cdot 32}{2} = 0.0096$$

$$\begin{aligned} P &= 1 - (0 + 0.0024 + 0.0096) \\ &= 1 - 0.012 \\ &= 0.988 \end{aligned}$$

$$\text{iii) } x=7, 4 \quad t=3$$

$$\begin{aligned} P &= 1 - P(x \leq 4) \\ &= 1 - (P(x=0) + P(x=1) + P(x=2) + P(x=3)) \end{aligned}$$

$$x=1 \quad t=3$$

$$e^{-6} = 0.002$$

$$\frac{e^{-2(3)} (2(3))^1}{1!} = \frac{e^{-6} (6)}{1} = 0.002(6) = 0.012$$

$$x=2 \quad t=3$$

$$\frac{e^{-2(3)} (2(3))^2}{2!} = \frac{e^{-6} (18)}{2} = 0.002(18) = 0.036$$

$$x=3 \quad t=3$$

$$\frac{e^{-2(3)} (2(3))^3}{3!} = \frac{e^{-6} (36)}{6} = 0.002(36) = 0.072$$

$$\begin{aligned} P &= 1 - (0.012 + 0.036 + 0.072) \\ &= 1 - 0.12 \\ &= 0.88 \end{aligned}$$

2 C

Rank correlation

 $n=10$

$R(X)-R(Y)$					
X	Y	$R(X)$	$R(Y)$	D	D^2
10	30	10	10	0	0
15	42	6	3	3	9
12	45	9	2	7	49
17	46	4	1	3	9
13	33	8	9	-1	1
16	34	5	8	-3	9
24	40	1	4	-3	9
14	35	7	7	0	0
22	39	2	5	-3	9
20	38	3	6	-3	9
					104

$$r = 1 - \frac{6 \cdot \sum D^2}{n \cdot (n^2 - 1)}$$

$$= 1 - \frac{6 \cdot 104}{10 \cdot (10^2 - 1)} = 1 - \frac{6 \cdot 104}{10 \cdot 99}$$

$$= 1 - \frac{104}{165} = \frac{61}{165}$$

4 A

X	2	3	4	5	6
$f(X)$	0.01	0.25	0.4	0.3	0.04

Mean? Variance?

$$\mu = \sum X_i \cdot f(X_i)$$

$$= 2(0.01) + 3(0.25) + 4(0.4) + 5(0.3) + 6(0.04)$$

$$= 0.02 + 0.75 + 1.6 + 1.5 + 0.24$$

$$\mu = 4.11$$

$$\text{Var} = \sum (X_i)^2 \cdot f(X_i) - \mu^2$$

$$= 2^2(0.01) + 3^2(0.25) + 4^2(0.4) + 5^2(0.3) + 6^2(0.04) - (4.11)^2$$

$$= 0.04 + 2.25 + 6.4 + 7.5 + 1.44 - 16.89$$

$$= 17.63 - 16.89 = 0.74$$

4 B

Poisson

$$\lambda = 5 \quad t = 1 \text{ (year)} \quad \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

i) $x \leq 3$

$$= P(x=0) + P(x=1) + P(x=2) + P(x=3)$$

$$x=1 \quad \frac{e^{-5} \cdot (5)^1}{1!} = e^{-5} (5) = 0.006 (5) = 0.03 \quad e^{-5} = 0.006$$

$$x=2 \quad \frac{e^{-5} \cdot (5)^2}{2!} = \frac{e^{-5} (25)}{2} = \frac{0.006 (25)}{2} = 0.075$$

$$x=3 \quad \frac{e^{-5} \cdot (5)^3}{3!} = \frac{e^{-5} (125)}{6} = \frac{0.006 (125)}{6} = 0.375$$

$$= 0.03 + 0.075 + 0.375$$

$$= 0.48$$

ii) $x \geq 1$
 $= 1 - P(x \leq 1)$

$$= 1 - (P(x=0) + P(x=1))$$

$$x=1$$

$$\frac{e^{-5} \cdot (5)^1}{1!} = e^{-5} 5 = 0.006 (5) = 0.03$$

$$1 - (0.03)$$

$$=$$

5 A

$$\mu = 10.28$$

$$\sigma = 0.12$$

$$Z = \frac{X - \mu}{\sigma}$$

i) $(10.00 < X < 10.50)$ $\rightarrow$ Formula used

$$= (-0.40 < Z < 1.82)$$

$$= (Z < 1.82) - (Z < -0.40)$$

$$= 0.96638 - 0.34458$$

$$= 0.62180$$

ii) $(X > 10.20)$

$$= 1 - (X < 10.20)$$

$$= 1 - (Z < -0.66)$$

$$= 1 - 0.25463 = 0.74537$$

iii) $(X \leq 10.35)$

$$= (Z < 0.58)$$

$$= 0.71904$$

5 B

$$18 / 5.4 / 20 / 46 / 25 / 48 / 5.3 / 27 / 26 / 37 / 40 / 36 / 42 / 25 / 2.7 / 3.3 / 2.8 / 40 / 45 / 45$$

$$n = 20$$

$$\text{Mean} = \frac{715}{20} = 35.75$$

$$\text{mode} = 45 \text{ repeated twice...}$$

$$\text{Median} = 36$$

if the median is 36 and mean is 35.75 which is similar that mean ~~36~~ it is almost averagely equal.

